

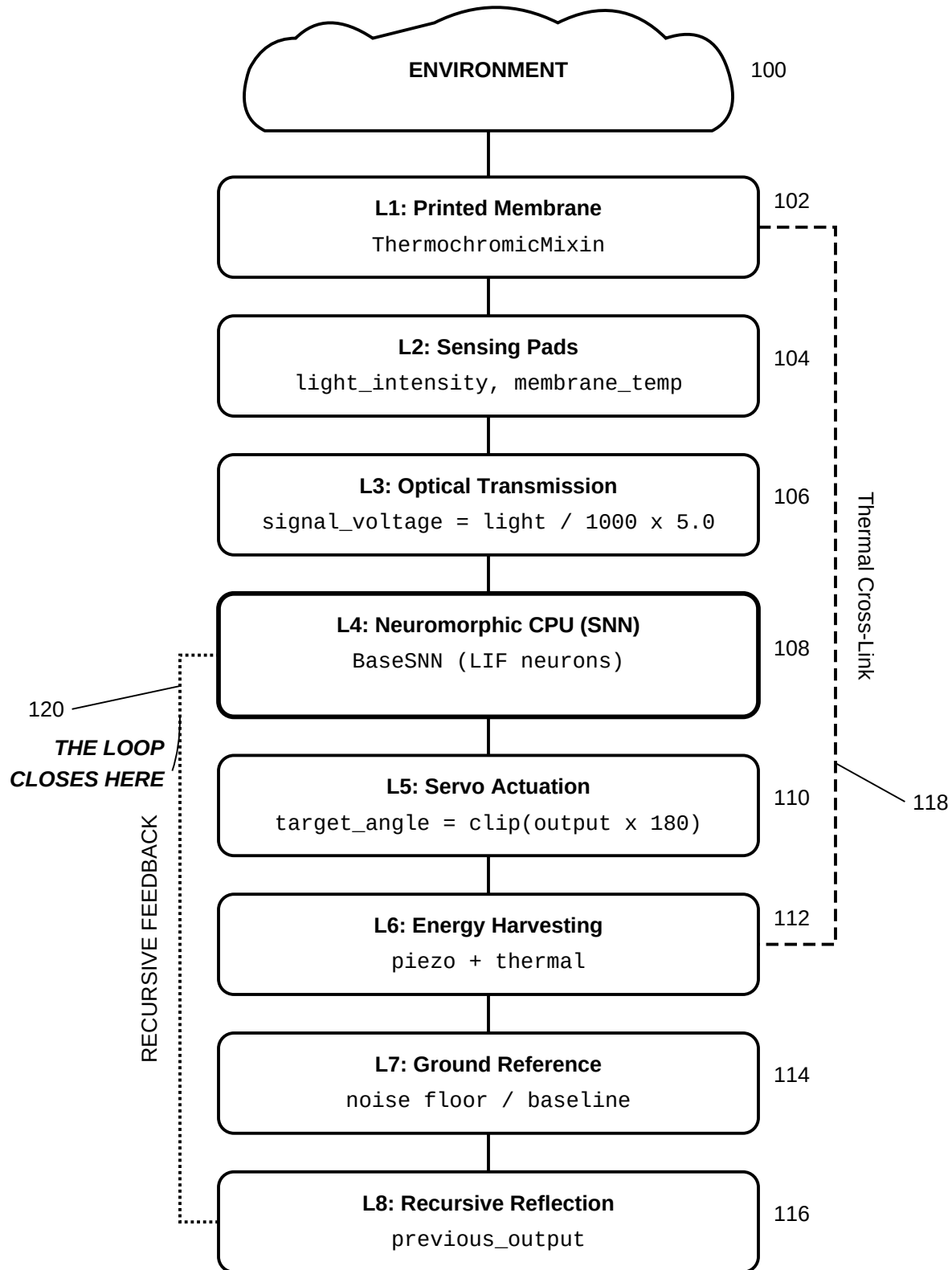


FIG. 1

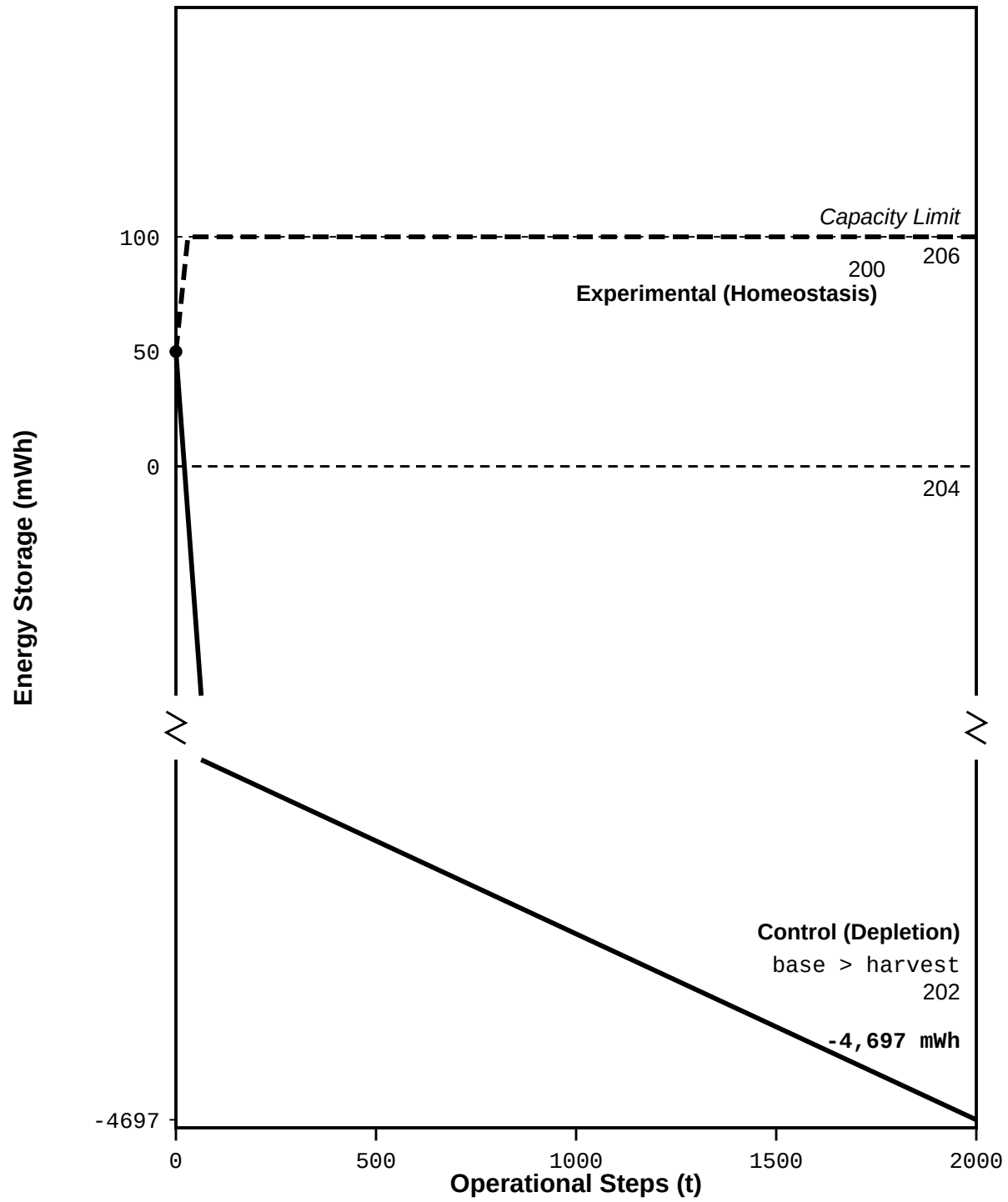
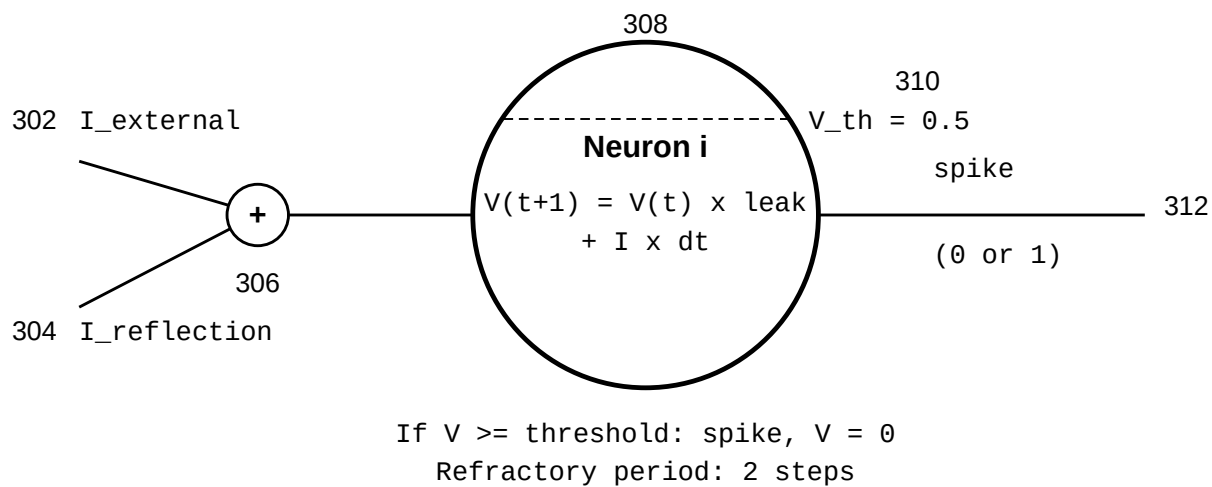
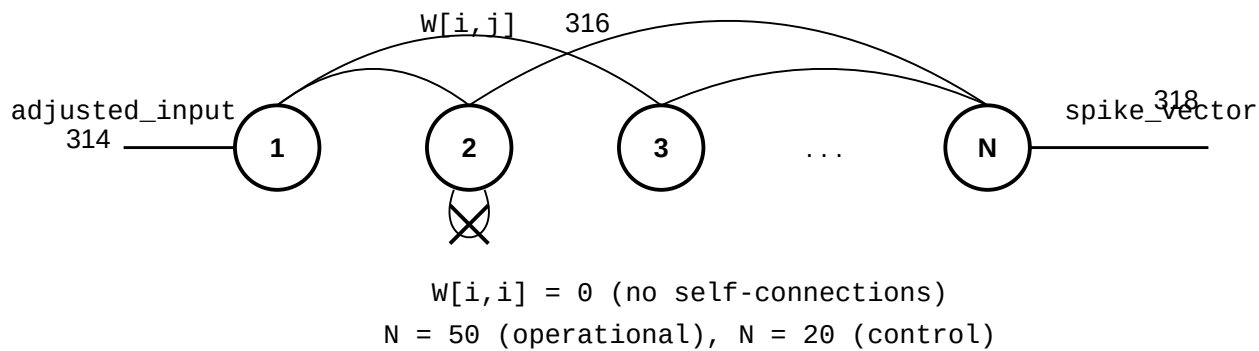


FIG. 2

Single LIF Neuron Detail



Network Topology

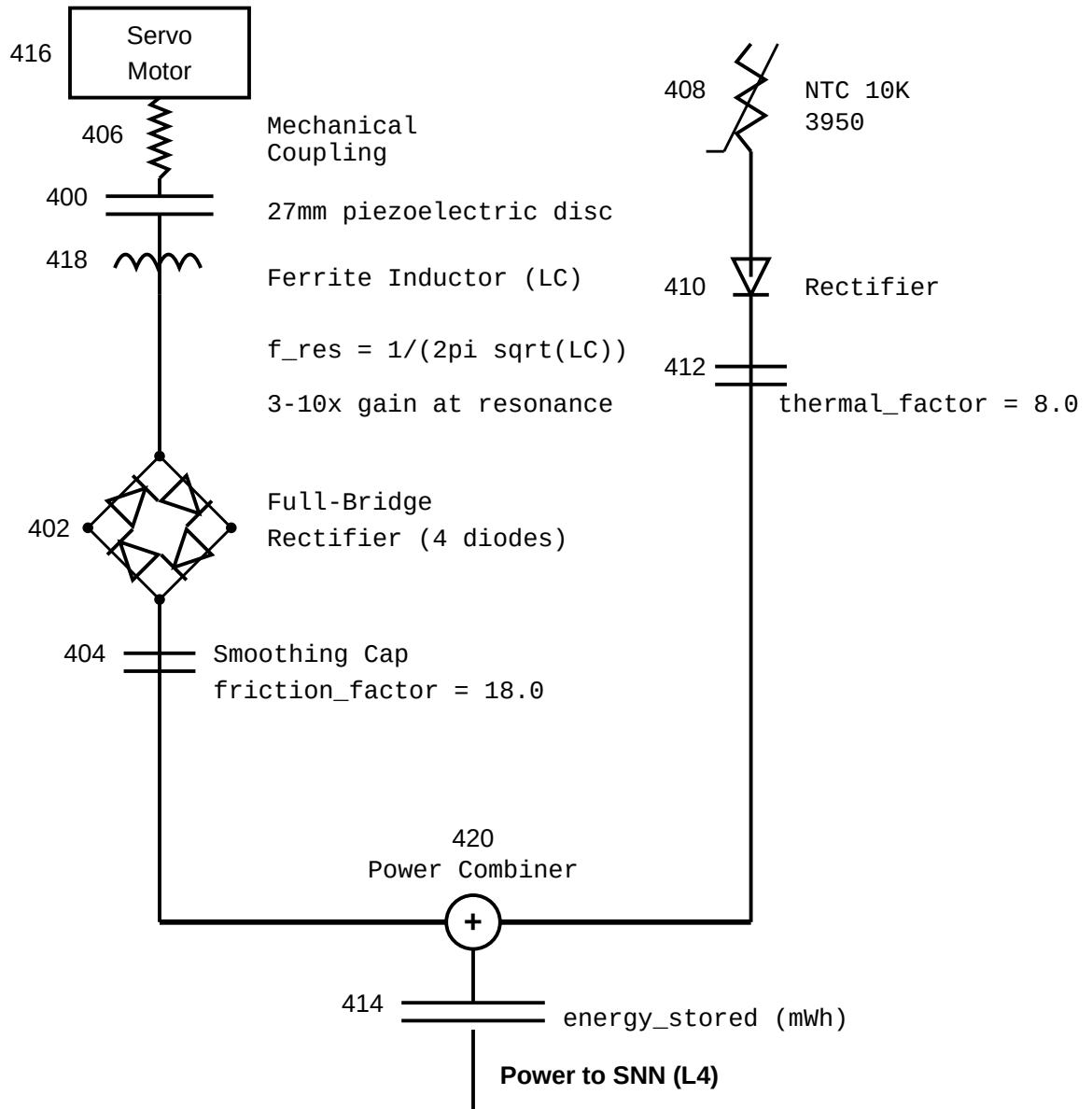


STDP Learning Rule

Post fires: $W \ += \ A+ \times \text{pre_trace}$ (LTP)
Pre fires: $W \ -= \ A- \times \text{post_trace}$ (LTD)
 $A- > A+$ (stability bias)

320

FIG. 3

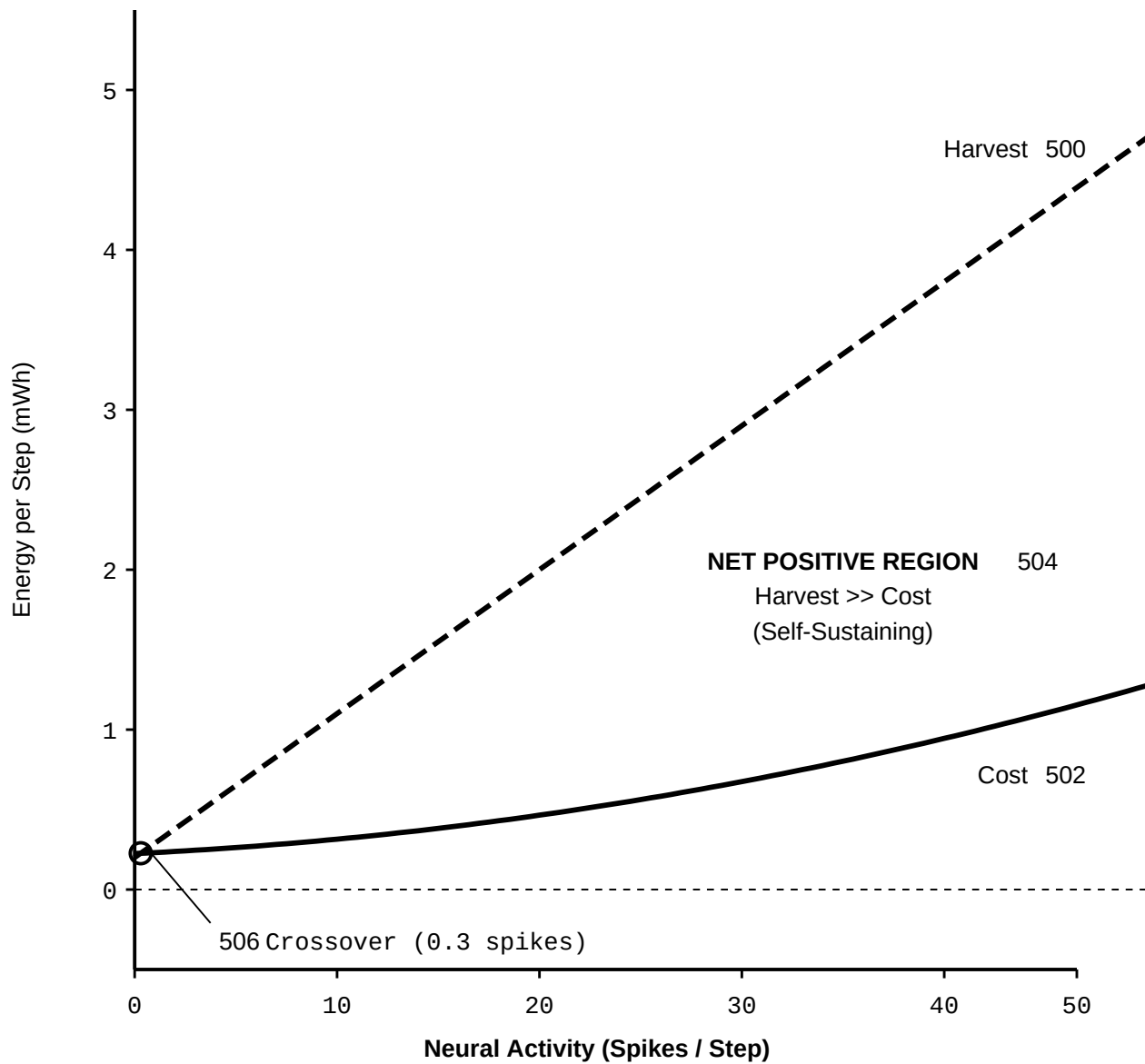
Piezoelectric Harvester**Thermoelectric Harvester****Power Budget**

Harvest = friction_harvest + thermal_harvest

Cost = base_consumption + activity_cost x spike_count²

Net = Harvest - Cost

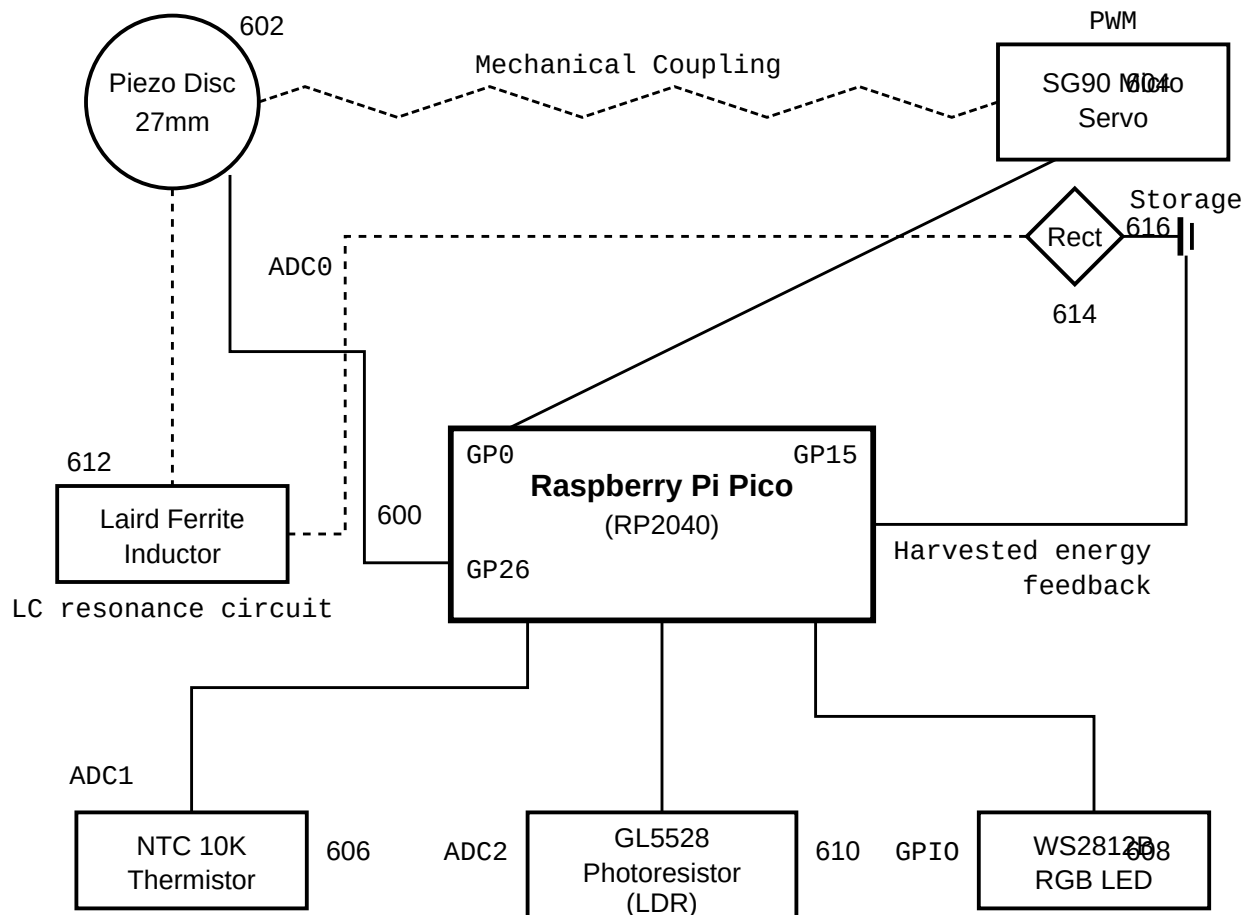
FIG. 4



$$\begin{aligned} \text{Harvest} &= 0.20 + (0.09 \times \text{spikes}) \text{ mWh/step} & 508 \\ \text{Cost} &= 0.225 + (0.006 \times s) + (0.0003 \times s^2) \text{ mWh/step} \\ \text{Net} &= \text{Harvest} - \text{Cost} \end{aligned}$$

Zone 1: Quiescent Drain
0 - 1 Spikes: Cost > Harvest

FIG. 5



Component Summary

- Raspberry Pi Pico (RP2040)
- SG90 Micro Servo
- 27mm Piezoelectric Disc
- NTC 10K Thermistor
- Photoresistor (LDR)
- WS2812B RGB LED
- Misc (wires, board, inductor)

———— Signal / Control

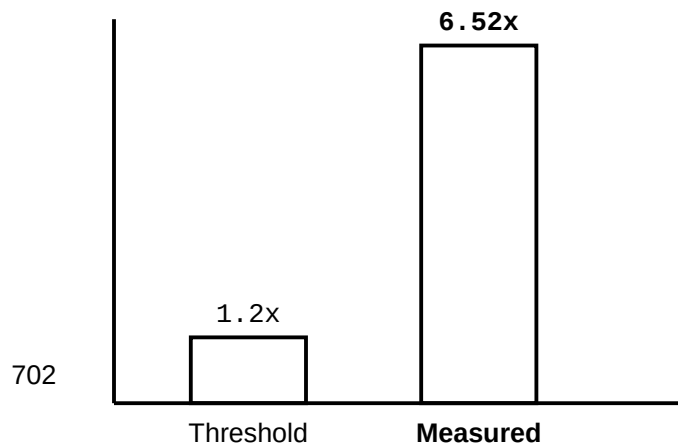
----- Energy Harvesting

FIG. 6

700

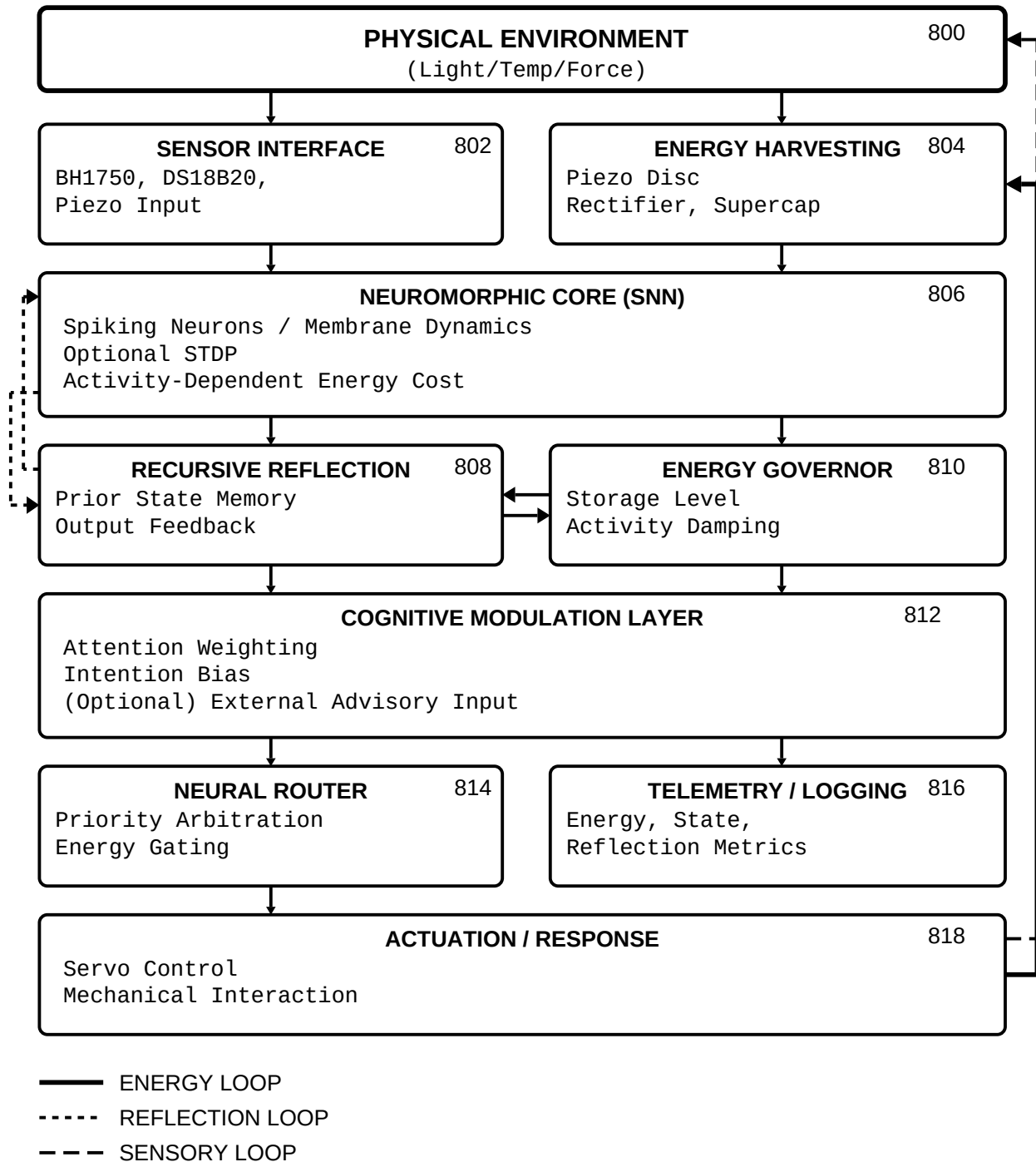
VALIDATION RESULTS SUMMARY

Claim	Criterion	Measured	Threshold	Result
Phase 7 Control Baseline (EnergyConfig)				
A	Closed-loop continuity	2000 pts	2000	PASS
B	Boundedness	bounded	bounded	PASS
C	Robustness (noise)	Std(theta) <= 45	<= 45	PASS
D	Input-output gain	corr >= 0.2	>= 0.2	PASS
E	Saturation ratio	sat <= 0.20	<= 0.20	PASS
Phase 7 Full Integration (Control confirms energy depletion)				
--	Energy at 2000 steps	-4,697 mWh	< 0	CONFIRMED
MCP Integration (BalancedEnergyConfig)				
--	Energy at 2000 steps	100.0 mWh (capped)	> 0	CONFIRMED
Phase 8 STDP				
F8.1	Weight entropy decrease	2.95 to 2.02 bits	decrease	PASS
F8.2	STDP MI > 1.2x frozen	6.52x	> 1.20	PASS
F8.3	Weight convergence	0.00%	< 10%	PASS

STDP Learning Efficacy (Claim F8.2)

All claims validated. System is reproducible via git tags.

FIG. 7



All computation is constrained by real-time energy availability derived from mechanical interaction.

FIG. 8